

Bayesian Networks for early detection of Coronary Artery Disease

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Abstract

Coronary artery disease (CAD) is one of the leading causes of death worldwide, yet its diagnosis often relies on costly and invasive procedures. This project explores Bayesian Networks as a interpretable computational alternative, estimating the probability of coronary artery disease from routine clinical measurements. We compare six network configurations on the Cleveland Heart Disease dataset (Detrano and others), from a fully data-driven baseline to models incorporating forbidden edges, forced edges grounded in medical literature, and latent variables, evaluating all through stratified 5-fold cross-validation. We then test cross-population generalization on the Indian Heart Disease dataset (Mendeley Data). The unconstrained model achieved the highest accuracy (0.87), while domain knowledge constraints improved interpretability without improving prediction. Generalization to the Indian dataset proved limited (0.54 accuracy), highlighting the impact of population-specific factors on transferability.

Introduction

Domain

The clinical diagnosis of CAD relies on a combination of risk factors, age, sex, chest pain type, resting blood pressure, serum cholesterol, and electrocardiographic findings, whose interactions are inherently probabilistic and often non-linear. Bayesian Networks are a natural modeling choice for this domain, given their ability to handle uncertainty, encode causal relationships, and support inference under partial evidence.

Aim

The purpose of this project is to explore how progressively incorporating domain knowledge into a learned Bayesian Network affects both predictive accuracy and clinical interpretability. Starting from a fully data-driven baseline, we incrementally constrain the structure learning process by introducing forbidden edges, forced edges grounded in medical literature (Santos and others), and latent variables representing unobserved clinical constructs such as overall coronary health and metabolic risk. We also assess whether the best-performing model trained on the Cleveland dataset generalizes to a structurally different population, namely the Indian Heart Disease dataset, which presents distinct demographic and epidemiological characteristics.

Method

We used Python with the `pgmpy` library for structure learning, parameter estimation, and inference. Continuous variables (age, `trestbps`, `chol`, `thalach`, `oldpeak`) were discretized according to clinically established thresholds: NCEP ATP III guidelines for cholesterol (NCEP Expert Panel), AHA classifications for blood pressure (Whelton and others), AHA target heart rate standards for maximum heart rate, GBD study protocols for age grouping, and diagnostic criteria for ST depression (Yap and others).

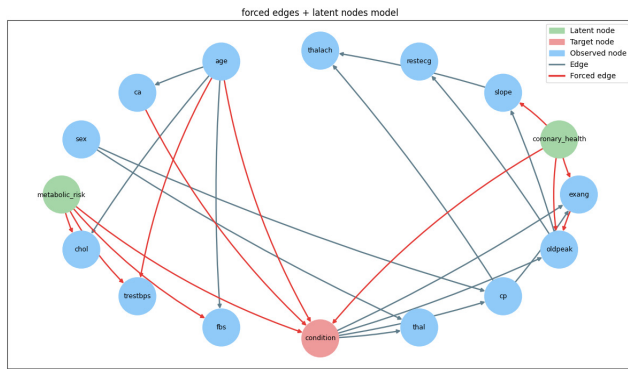
Structure learning was performed via Hill Climbing with AIC scoring, progressively augmented with:

1. Forbidden edges to prevent clinically implausible causal directions (e.g. `condition` $\rightarrow$ `age`, `condition` $\rightarrow$ `sex`);
2. Forced edges justified by peer-reviewed studies (Santos and others) (e.g., `age` $\rightarrow$ `condition`, `ca` $\rightarrow$ `condition`, `oldpeak` $\rightarrow$ `condition`);
3. Latent variables (`coronary_health`, `metabolic_risk`) introduced manually based on domain knowledge, with parameters estimated via Expectation-Maximization.

Parameters for all observed-only models were estimated using a Bayesian Estimator with BDeu prior. Inference was performed via Variable Elimination. All models were evaluated on accuracy and recall using stratified 5-fold cross-validation.

Results

The unconstrained model achieved the highest accuracy on the Cleveland dataset (0.87), suggesting that purely data-driven structure learning is competitive when training and test data come from the same population. Incorporating domain knowledge through forbidden edges, forced edges, and latent variables did not improve predictive performance, but produced more interpretable network structures whose edges align with established clinical relationships. Cross-dataset generalization proved limited: a model trained on Cleveland dropped to 0.54 accuracy on the Indian dataset, indicating that population-specific factors, such as demographic differences and distinct risk factor distributions, substantially affect network transferability.



Model

The network nodes correspond to the 13 clinical features of the Cleveland dataset plus the binary target variable `condition` (0 = no disease, 1 = disease). Continuous variables were discretized into clinically meaningful categories prior to learning (see Method). The two latent nodes, `coronary_health` and `metabolic_risk`, are binary hidden variables with no corresponding observed measurement; their states are estimated via EM.

We built four progressively constrained models. The **unconstrained model** learns structure freely from data via Hill Climbing with AIC scoring. The **blacklist model** adds forbidden edges to prevent clinically implausible directions (e.g., `condition` → `age`, `condition` → `sex`). The **forced edges model** additionally requires edges grounded in medical literature. The **latent node model** introduces `coronary_health` as a hidden common cause of `oldpeak`, `exang`, `slope`, and `condition`, with structure among observed variables still learned via Hill Climbing and CPDs estimated via EM.

Analysis

Experimental setup

We evaluated six models on the Cleveland dataset using stratified 5-fold cross-validation, measuring accuracy and recall. Recall was included given the clinical cost of false negatives in CAD diagnosis. We expected the unconstrained model to achieve the highest accuracy, while domain knowledge constraints were expected to trade off predictive performance for interpretability. For the cross-dataset experiment, we applied the best-performing latent model to the Indian dataset, expecting a performance drop due to demographic differences between the two populations.

Results

Table summarizes the performance of all models on the Cleveland dataset via stratified 5-fold cross-validation. The unconstrained model achieved the highest accuracy (0.87), confirming that purely data-driven learning is competitive on in-distribution data. Incorporating domain knowledge did not improve predictive performance but produced more interpretable structures. Latent variable models underperformed, likely due to overfitting on the small dataset (303

patients). In the cross-dataset experiment, the latent model trained on Cleveland dropped to 0.54 accuracy on the Indian dataset, while a model trained directly on Indian data reached 0.68, confirming that population-specific factors substantially affect transferability.

Model	Accuracy	Recall
Unconstrained	0.87	0.84
Blacklist	0.78	0.78
Forced edges	0.78	0.74
Latent (cor. health)	0.76	0.72
Latent (metab. risk)	0.79	0.71
Latent (forced+latents)	0.73	0.67
Cleveland → Indian	0.54	0.57
Indian (own data)	0.68	0.80

Conclusion

This project showed that incorporating medical domain knowledge into Bayesian Network structure learning improves interpretability but does not consistently improve predictive accuracy, at least on a dataset as small as Cleveland. The unconstrained model remained the strongest predictor, likely because AIC scoring already penalizes overly complex structures. The main limitation of our study is dataset size: with 303 patients, the learned CPDs are noisy and latent variable models are particularly prone to overfitting. Cross-dataset generalization proved limited, highlighting that Bayesian Networks trained on a specific population should be applied with caution to demographically different cohorts. Future work could explore larger datasets, richer latent variable structures, or transfer learning approaches to improve cross-population robustness.

Links to external resources

The project code, notebooks and datasets are available on GitHub.

References

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